



Lean production

The goal of lean production is to reduce the resources used to supply goods and services to consumers. By cutting down human effort, materials, space, capital, and time, lean production cuts costs.

How it works

The focus in lean production is on efficiency to maximize value for the customer, but without affecting quality. Lean seeks to eliminate all activities that do not add value to the production process, including holding inventory (stock), repairing faults, and unnecessary movement of people and products around a manufacturing plant.

Optimizing the flow of products and services through value streams—sequences of activity that flow horizontally across technologies, assets, and functions to customers—allows the business to respond more quickly to consumer demand. Efficiency also makes it simpler and more accurate to manage information.

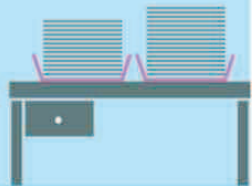


NEED TO KNOW

- **Kanban cards** Toyota's demand-driven scheduling triggers
- **Value-stream mapping** The process of analyzing how the product gets from start to finish now and the design of an improved flow for the future

Case study: how Toyota eliminates waste

Lean production is about getting rid of waste, sometimes called non-value-added activities. Car manufacturer Toyota has identified eight areas of waste and a lean approach to counter these.

Overproduction	Waste Items produced surplus to customer demand 	Waiting	Waste Unproductive time spent waiting for material, information, equipment, tools 
	Lean solution Manufacture based upon a pull system, producing products as customers order them 		Lean solution All resources provided on a just-in-time (JIT) basis—not too early, not too late (see pp.290–291) 
Excess inventory	Waste Extra, unwanted stock held in inventory 	Defects	Waste Consumes materials and uses up labor; results in customer complaints 
	Lean solution Kanban cards used to indicate material order points: how much, from where, and to where 		Lean solution Total quality management (see pp.292–293) used to improve all areas 